

PLAYING TOUGHER FOR ONE'S SIZE: POSITIONAL PHYSICALITY IN THE NHL

RENTO SAIJO¹, SIAN LEE², YEONGJUN SEO³, JEFF MOHER¹

¹ Connecticut College; ² Michigan State University; ³ University of Tennessee, Knoxville

Introduction

Adding size to a roster does not guarantee the physical presence a team needs. That presence emerges through contact, puck battles, and contested space. We develop calibrated size above expected (CSAx) to quantify playing tougher for one's size and examine what that profile means for NHL roles and continued participation.

Methods

We combine NHL events with All Three Zones' manually tracked puck plays from 2021-22 through 2024-25, covering 1,485 forward-seasons and 829 defenseman-seasons. Separate positional ridge regressions estimate the body size implied by direct contact and position-specific indirect physicality. We calibrate held-out predictions against listed height and weight, then standardize residuals within position and season. Higher CSAx indicates more physical presence than expected for that frame. Playoff comparisons use regular-season baselines excluded from score construction.

Results

Earlier scouting reports offer independent support: players described as actively physical subsequently receive higher CSAx in both positions (Figure 1).

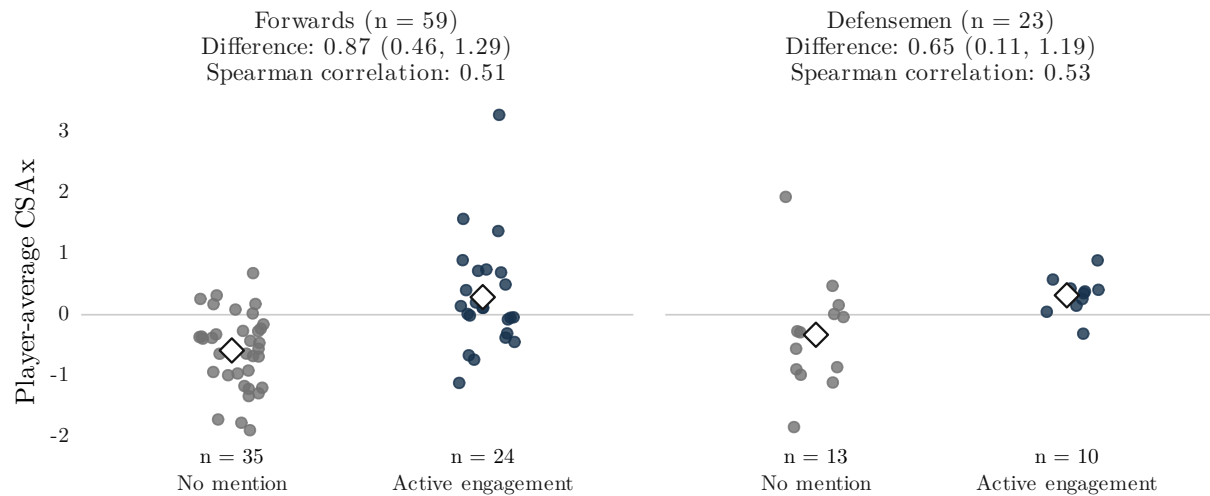


Figure 1. Player-average CSAx by scouting description. Points represent players; diamonds mark means. Differences show 95% intervals. No mention does not establish soft play.

Across the full sample, higher-scoring players spend less of their ice time on power plays and more killing penalties (Table 1). When play shifts to the postseason, higher-scoring forwards show smaller proportional increases in delivered hits over each team's first four playoff games; defensive differences remain uncertain. Looking ahead, the adjusted probability of reaching 300 NHL minutes next season is 2.92 percentage points higher for forwards and 4.23 for defensemen at CSAx +1 versus -1. Among returners, however, minutes per game show no clear increase.

Table 1. Adjusted CSAx +1 versus -1 comparisons (two standard deviations apart; 95% intervals in parentheses). pp means percentage points; ice time is next-season minutes per game among returners. The hit-rate ratio compares playoff-to-baseline rate changes; below one means a smaller proportional increase.

Outcome	Forwards	Defensemen
Power-play share (pp)	-1.22 (-1.78, -0.67)	-0.88 (-1.51, -0.25)
Penalty-kill share (pp)	1.40 (0.74, 2.06)	1.31 (0.74, 1.89)
Playoff hit-rate change (ratio)	0.78 (0.69, 0.88)	0.90 (0.77, 1.05)
Next-season continuation (pp)	2.92 (0.03, 5.81)	4.23 (0.60, 7.86)
Ice time (min/game)	-0.18 (-0.36, 0.00)	-0.14 (-0.43, 0.14)

Conclusion

CSAx connects a familiar scouting judgment to measurable differences in deployment, playoff behavior, and NHL participation. Selective tracking, one scouting rater, and weak 2024-25 defensive size prediction limit generalization. For teams, CSAx broadens physicality evaluation beyond size and hit counts, helping identify players who bring more physical presence than their frame alone would suggest.